

Day 3: Conditional Statements in Python (Detailed Notes)

WHAT ARE CONDITIONAL STATEMENTS?

Conditional statements help Python make decisions.

They allow your program to execute different code based on conditions.

REAL-LIFE EXAMPLE

Suppose:

- If salary > 50,000 → Eligible for Bonus
- Otherwise → Not Eligible

Python checks the condition and takes action accordingly.

Why Are Conditional Statements Important?

Used in:

- Login Systems
 - ATM Machines
 - Employee Bonus Calculations
 - Sales Targets
 - Student Results
 - Dashboard KPI Monitoring
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Types of Conditional Statements

1. if
 2. if-else
 3. if-elif-else
 4. Nested if
-

1. if Statement

Syntax:

```
if condition:  
    statement
```

EXAMPLE

```
salary = 60000

if salary > 50000:
    print("Bonus Eligible")
```

Output:

```
Bonus Eligible
```

FLOW

```
Condition True
    ↓
Execute Code

Condition False
    ↓
Skip Code
```

EXAMPLE 2

```
age = 20

if age >= 18:
    print("You Can Vote")
```

Output:

```
You Can Vote
```

2. if-else Statement

Used when there are two possible outcomes.

Syntax:

```
if condition:
    statement
else:
    statement
```

EXAMPLE

```
age = 16

if age >= 18:
    print("Eligible")
else:
    print("Not Eligible")
```

Output:

```
Not Eligible
```

EVEN OR ODD NUMBER

```
number = int(input("Enter Number: "))

if number % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Business Example

Employee Bonus

```
salary = 55000

if salary > 50000:
    print("Bonus Approved")
else:
    print("Bonus Not Approved")
```

3. if-elif-else Statement

Used when multiple conditions exist.

Syntax:

```
if condition:
    statement
```

```
elif condition:
    statement

else:
    statement
```

EXAMPLE: STUDENT GRADE

```
marks = 85

if marks >= 90:
    print("A Grade")

elif marks >= 75:
    print("B Grade")

elif marks >= 60:
    print("C Grade")

else:
    print("Fail")
```

Output:

```
B Grade
```

SALES PERFORMANCE EXAMPLE

```
sales = 120000

if sales >= 150000:
    print("Excellent")

elif sales >= 100000:
    print("Good")

elif sales >= 50000:
    print("Average")

else:
    print("Poor")
```

Output:

```
Good
```

4. Nested if Statement

An if inside another if.

Syntax:

```
if condition:

    if condition:
        statement
```

EXAMPLE

```
salary = 60000
experience = 4

if salary > 50000:

    if experience >= 3:
        print("Promotion Eligible")
```

Output:

```
Promotion Eligible
```

Logical Operators with Conditions

AND

Both conditions must be True.

```
salary = 60000
experience = 5

if salary > 50000 and experience >= 3:
    print("Promotion Eligible")
```

OR

At least one condition must be True.

```
salary = 25000
experience = 5

if salary > 50000 or experience >= 5:
    print("Interview Eligible")
```

NOT

Reverses the result.

```
is_blocked = False

if not is_blocked:
    print("Access Granted")
```

Real Data Analyst Examples

KPI ACHIEVEMENT

```
sales = 120000
target = 100000

if sales >= target:
    print("Target Achieved")
else:
    print("Target Not Achieved")
```

PROFIT CHECK

```
sales = 100000
cost = 70000

profit = sales - cost

if profit > 0:
    print("Profit")
else:
    print("Loss")
```

DATA QUALITY CHECK

```
records = 1000

if records > 0:
    print("Data Available")
else:
    print("No Data")
```

Mini Project 1: Voting Eligibility

```
age = int(input("Enter Age: "))

if age >= 18:
    print("Eligible to Vote")
else:
    print("Not Eligible")
```

Mini Project 2: Grade Calculator

```
marks = float(input("Enter Marks: "))

if marks >= 90:
    print("A")

elif marks >= 75:
    print("B")

elif marks >= 60:
    print("C")

else:
    print("Fail")
```

Mini Project 3: Employee Bonus System

```
salary = float(input("Enter Salary: "))

if salary >= 50000:
    bonus = salary * 0.10
```

```
else:  
    bonus = salary * 0.05  
  
print("Bonus =", bonus)
```

Mini Project 4: Profit or Loss

```
sales = float(input("Enter Sales: "))  
cost = float(input("Enter Cost: "))  
  
profit = sales - cost  
  
if profit > 0:  
    print("Profit =", profit)  
  
elif profit < 0:  
    print("Loss =", abs(profit))  
  
else:  
    print("No Profit No Loss")
```

Practice Questions

BASIC

1. Check if age is greater than 18.
 2. Check if salary is above 30,000.
 3. Check if marks are above 50.
 4. Check whether a number is positive.
 5. Check whether a number is negative.
-

IF-ELSE

6. Check even or odd.
 7. Check pass or fail.
 8. Check profit or loss.
 9. Check target achieved or not.
 10. Check adult or minor.
-

IF-ELIF-ELSE

11. Create grade system.
 12. Create sales performance system.
 13. Create employee rating system.
 14. Create income tax slab system.
 15. Create attendance grading system.
-

NESTED IF

16. Promotion eligibility.
 17. Loan approval system.
 18. Scholarship eligibility.
 19. Job eligibility.
 20. Bonus eligibility.
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BUSINESS PROBLEMS

21. Calculate employee bonus.
 22. Check sales target.
 23. Check inventory stock.
 24. Check customer discount eligibility.
 25. Check profit margin level.
-

Assignment (Data Analyst Level)

Create a Sales KPI Program:

```
sales = float(input("Enter Sales: "))  
target = float(input("Enter Target: "))
```

Conditions:

- Sales $\geq$ 120% Target → Excellent
- Sales $\geq$ 100% Target → Good
- Sales $\geq$ 80% Target → Average
- Otherwise → Poor

Expected Output:

```
Sales: 120000  
Target: 100000
```

Interview Questions

1. What is a conditional statement?
2. Difference between if and if-else?
3. What is elif?
4. What is nested if?
5. What is the use of logical operators?
6. Difference between AND and OR?
7. When should nested if be used?
8. What happens if all conditions are False?
9. What is indentation in Python?
10. Give a real business example of conditional statements.